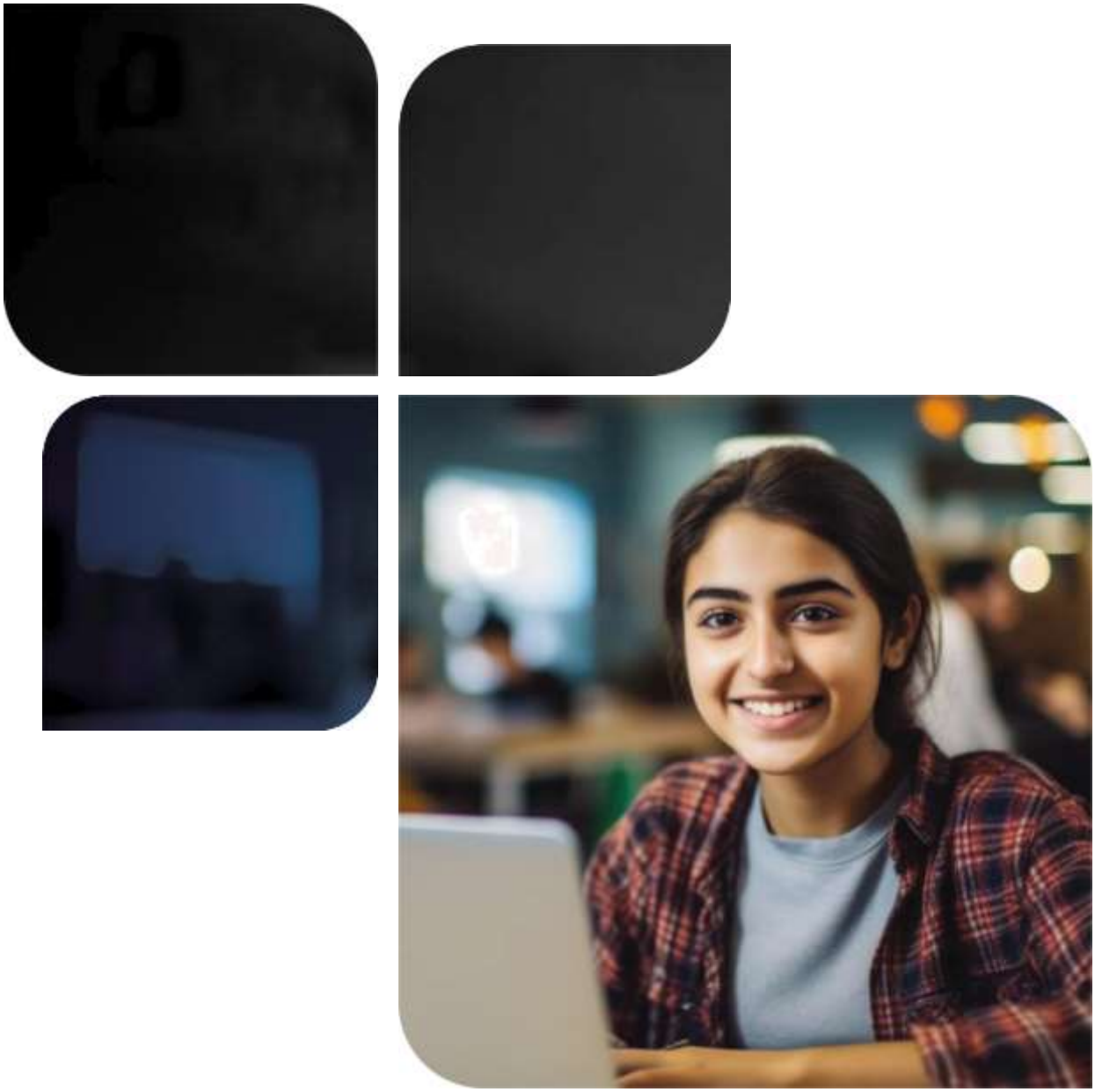
GLOBAL QUEST
TECHNOLOGIES

An ISO 9001 & ISO 2100a
Certified Organizaton



The Quest
For your Dream Job
Ends Here!!

www.gqtech.in

Project Report On
Java Quiz Application

Submitted by:
Pharangi Harsha Vardhan
Date: 19/01/2026

1. PROBLEM STATEMENT

The objective of this project is to design and implement a console-based KBC (Kaun Banega Crorepati) quiz application using Java, where a player answers multiple-choice questions, earns rewards for correct answers, uses limited lifelines, and exits the game immediately upon selecting a wrong answer. The system should be modular, user-friendly, and suitable for demonstrating object-oriented programming concepts.

2. OBJECTIVES OF THE PROJECT

The objectives of this project are:

- To design a modular console-based application.
- To implement conditional logic and control flow.
- To simulate a real-time quiz game environment.
- To understand class interaction and code reusability.

3. PROJECT DESCRIPTION

The Java Based KBC Application is a menu-driven console program. At the start of the game, the player enters basic details such as name and age. The game then presents a total of 10 questions, each having four options.

The rules of the game are:

- The player must answer the current question correctly to move to the next question.
- Each correct answer provides a predefined reward amount.
- If the player selects a wrong answer at any stage, the game ends immediately.
- Two lifelines are available: 50-50 and Audience Poll.
- Each lifeline can be used only once.
- Lifelines are available only for questions 1 to 9 and is not allowed for the 10th question.
- Correct answers are displayed in green color and wrong answers in red color.

4. CONCEPTS USED

1. Classes and Objects – Used to structure the program and run the KBC game.
2. Arrays – Used to store questions, options, correct answers, and prizes.
3. Loops – Used to display questions and control the game flow.
4. Conditional Statements – Used to check answers and decide results.
5. Scanner & Random Functions – Scanner is used for user input and random for lifelines.

5. CODE IMPLEMENTATION

```java
import java.util.Scanner;

public class SimpleKBC {
    public static void main(String[] args) {

        String[] questions = {
                "Which country is the largest by land area?",
                "What is the capital of France?",
                "Which organization has 193 member states?",
                "Which country is not a permanent UN Security Council member?",
                "What is the currency of Japan?",
                "Which country is the world's largest democracy?",
                "What is the capital of Canada?",
                "Which country has the world's largest population?",
                "What is the official language of Brazil?",
                "Which country is not part of the European Union?"
        };

        String[][] options = {
                { "China", "Russia", "Canada", "USA" },
                { "London", "Berlin", "Paris", "Madrid" },
                { "NATO", "UNESCO", "United Nations", "European Union" },
                { "China", "France", "Germany", "Russia" },
                { "Yen", "Won", "Yuan", "Baht" },
                { "USA", "India", "Indonesia", "Brazil" },
                { "Toronto", "Vancouver", "Ottawa", "Montreal" },
                { "India", "China", "USA", "Indonesia" },
                { "Spanish", "Portuguese", "French", "English" },
                { "Germany", "France", "Norway", "Italy" }
        };

        int[] correctAnswers = { 1, 2, 2, 2, 0, 1, 2, 1, 1, 2 };
        int[] prizes = { 1000, 2000, 3000, 5000, 10000, 20000, 40000, 80000, 160000, 320000 };

        boolean fiftyFiftyAvailable = true;
        boolean audiencePollAvailable = true;
        int totalPrize = 0;
        Scanner scanner = new Scanner(System.in);

        System.out.println("==================================");
        System.out.println("       KAUN BANEGA CROREPATI ");
        System.out.println("==================================");

        System.out.print("Enter your name: ");
        String playerName = scanner.nextLine();
```

```java
        System.out.println("==================================");
        System.out.println("       KAUN BANEGA CROREPATI ");
        System.out.println("==================================");

        System.out.print("Enter your name: ");
        String playerName = scanner.nextLine();

        System.out.println("\nWelcome, " + playerName + "! Let's begin the game!");
        System.out.println("You have 2 lifelines: 50-50 and Audience Poll");
        System.out.println("You can use each lifeline only once during the game.\n");

        for (int i = 0; i < questions.length; i++) {
            System.out.println("\nQuestion " + (i + 1) + " for Rs." + prizes[i] + ":");
            System.out.println(questions[i]);

            for (int j = 0; j < 4; j++) {
                System.out.println((j + 1) + ". " + options[i][j]);
            }

            if ((fiftyFiftyAvailable || audiencePollAvailable) && i < questions.length - 1) {
                System.out.println("5. Use Lifeline");
            }

            System.out.print("\nEnter your choice (1-5): ");
            int choice = scanner.nextInt();

            if (choice == 5) {
                if ((fiftyFiftyAvailable || audiencePollAvailable) && i < questions.length - 1) {

                    System.out.println("\nAvailable Lifelines:");
                    if (fiftyFiftyAvailable)
                        System.out.println("1. 50-50");
                    if (audiencePollAvailable)
                        System.out.println("2. Audience Poll");
                    System.out.print("Select lifeline (0 to go back): ");

                    int lifelineChoice = scanner.nextInt();

                    if (lifelineChoice == 1 && fiftyFiftyAvailable) {

                        System.out.println("\n50-50 Lifeline:");
                        int correct = correctAnswers[i];
                        int wrong1, wrong2;

                        do {
```

```java
                    do {
                        wrong1 = (int) (Math.random() * 4);
                    } while (wrong1 == correct);

                    do {
                        wrong2 = (int) (Math.random() * 4);
                    } while (wrong2 == correct || wrong2 == wrong1);

                    for (int j = 0; j < 4; j++) {
                        if (j == wrong1 || j == wrong2) {
                            System.out.println((j + 1) + ". [HIDDEN]");
                        } else {
                            System.out.println((j + 1) + ". " + options[i][j]);
                        }
                    }

                    fiftyFiftyAvailable = false;
                    i--;
                    continue;
                } else if (lifelineChoice == 2 && audiencePollAvailable) {

                    System.out.println("\nAudience Poll Results:");
                    int correct = correctAnswers[i];
                    int correctPercent = 60 + (int) (Math.random() * 31); // 60-90%
                    int remaining = 100 - correctPercent;

                    int[] percents = new int[4];
                    percents[correct] = correctPercent;

                    for (int j = 0; j < 4; j++) {
                        if (j != correct) {
                            percents[j] = (int) (Math.random() * (remaining / 3));
                            remaining -= percents[j];
                        }
                    }

                    percents[0] += remaining;

                    for (int j = 0; j < 4; j++) {
                        System.out.println((j + 1) + ". " + options[i][j] + ": " + percents[j] + "%");
                    }

                    audiencePollAvailable = false;
                    i--;
```

```java
                            percents[j] = (int) (Math.random() * (remaining / 3));
                            remaining -= percents[j];
                        }
                    }

                    percents[0] += remaining;

                    for (int j = 0; j < 4; j++) {
                        System.out.println((j + 1) + ". " + options[i][j] + ": " + percents[j] + "%");
                    }

                    audiencePollAvailable = false;
                    i--;
                    continue;
                }
            } else {
                if (i == questions.length - 1) {
                    System.out.println("\n\033[36mCannot use lifelines on the last question!\033[0m");
                } else {
                    System.out.println("No lifelines available!");
                }
                i--;
                continue;
            }
        }

        if (choice - 1 == correctAnswers[i]) {
            totalPrize = prizes[i];
            System.out.println("\n\033[32mCorrect Answer!\033[0m");
            System.out.println("Congratulations! You've won Rs. " + totalPrize + " so far!");
        } else {
            System.out.println("\n\033[31mWrong Answer!\033[0m");
            System.out.println("Game Over! Your total prize money: Rs. " + totalPrize);
            break;
        }

        if (i == questions.length - 1) {
            System.out.println("\n\033[36mCONGRATULATIONS " + playerName.toUpperCase() + "!\033[0m");
            System.out.println("You've answered all questions correctly and won Rs. " + totalPrize + "!");
        }
    }

    scanner.close();
}
}
```

6. OUTPUT : Starting the game

```
=====================================
        KAUN BANEGA CROREPATI
=====================================
Enter your name: Harsha

Welcome, Harsha! Let's begin the game!
You have 2 lifelines: 50-50 and Audience Poll
You can use each lifeline only once during the game.


Question 1 for Rs.1000:
Which country is the largest by land area?
1. China
2. Russia
3. Canada
4. USA
5. Use Lifeline

Enter your choice (1-5): 2

Correct Answer!
Congratulations! You've won Rs. 1000 so far!

Question 2 for Rs.2000:
What is the capital of France?
1. London
2. Berlin
3. Paris
4. Madrid
5. Use Lifeline

Enter your choice (1-5): 3

Correct Answer!
Congratulations! You've won Rs. 2000 so far!
```

Using Lifelines:

```
Question 7 for Rs.40000:
What is the capital of Canada?
1. Toronto
2. Vancouver
3. Ottawa
4. Montreal
5. Use Lifeline

Enter your choice (1-5): 5

Available Lifelines:
1. 50-50
2. Audience Poll
Select lifeline (0 to go back): 2

Audience Poll Results:
1. Toronto: 16%
2. Vancouver: 0%
3. Ottawa: 84%
4. Montreal: 0%

Question 7 for Rs.40000:
What is the capital of Canada?
1. Toronto
2. Vancouver
3. Ottawa
4. Montreal
5. Use Lifeline

Enter your choice (1-5): 3

Correct Answer!
Congratulations! You've won Rs. 40000 so far!
```

Wrong answer:

```
Question 2 for Rs.2000:
What is the capital of France?
1. London
2. Berlin
3. Paris
4. Madrid
5. Use Lifeline

Enter your choice (1-5):
3

Correct Answer!
Congratulations! You've won Rs. 2000 so far!

Question 3 for Rs.3000:
Which organization has 193 member states?
1. NATO
2. UNESCO
3. United Nations
4. European Union
5. Use Lifeline

Enter your choice (1-5): 2

Wrong Answer!
Game Over! Your total prize money: Rs. 2000
```

Last Question and end of the Quiz:

```
Question 10 for Rs.320000:
Which country is not part of the European Union?
1. Germany
2. France
3. Norway
4. Italy

Enter your choice (1-5): 5

Cannot use lifelines on the last question!

Question 10 for Rs.320000:
Which country is not part of the European Union?
1. Germany
2. France
3. Norway
4. Italy

Enter your choice (1-5): 3

Correct Answer!
Congratulations! You've won Rs. 320000 so far!

CONGRATULATIONS HARSHA!
You've answered all questions correctly and won Rs. 320000!
```

7. CONCLUSION

The KBC Quiz project successfully demonstrates the implementation of a dynamic, current-affairs–based quiz system inspired by Kaun Banega Crorepati. By structuring questions, options, correct answers, and prize levels in a clean and scalable format, the project ensures ease of maintenance and future expansion. Incorporating real-time and recent events enhances user engagement while also promoting awareness of global news, sports, technology, and entertainment. Overall, this project strengthens core programming concepts such as arrays, indexing, and logical flow, while delivering an interactive and educational quiz experience that can be further extended with lifelines, timers, score tracking, and a user-friendly interface.

GLOBAL QUEST
TECHNOLOGIES

fuel your
passion for
IT with
our **guidance.**

Global Quest Technologies

#324,2nd Floor, 3 A Cross, Near
Seshadripuram First Grade College,
Above City Union Bank,
Yelahanka New Town,
Bengaluru-560064

+91 9448 403 469 | 080-49720009
info@gqtech.in | www.gqtech.in